

301. Photographic films and plates have an essential ingredient of

- (a) Silver nitrate
- (b) Silver bromide
- (c) Sodium chloride
- (d) Oleic acid

Answer: B — (b) Silver bromide

Chemical reason: Photographic emulsions traditionally contain AgBr crystals dispersed in gelatin because silver bromide is strongly photosensitive.

302. Which of the following does not react with AgCl

- (a) NaNO_3
- (b) Na_2CO_3
- (c) $\text{Na}_2\text{S}_2\text{O}_3$
- (d) NH_4OH

Answer: A — (a) NaNO_3

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $\text{S}_2\text{O}_3^{2-}$. Under the stated conditions this leads to NaNO_3 .

303. Which one of the following is known as lunar caustic when in fused state

- (a) Silver nitrate
- (b) Silver sulphate
- (c) Silver chloride
- (d) Sodium sulphate

Answer: A — (a) Silver nitrate

Chemical reason: Fused AgNO_3 is called lunar caustic. It was used as a cauterizing agent and the term “lunar” historically refers to silver.

304. Which silver halide is used in medicine

- (a) AgNO_3
- (b) AgCl
- (c) AgBr
- (d) AgF

Answer: A — (a) AgNO_3

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $\text{S}_2\text{O}_3^{2-}$. Under the stated conditions this leads to AgNO_3 .

305. When silver nitrate is heated to red hot, what is formed

- (a) Ag
- (b) Ag_2O
- (c) Ag_2O_3
- (d) AgO_2

Answer: A — (a) Ag

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $\text{S}_2\text{O}_3^{2-}$. Under the stated conditions this leads to Ag.

306. For making Ag from AgNO_3 , which of the following is used

- (a) PH_3
- (b) AsH_3
- (c) Na_2CO_3
- (d) NH_3

Answer: A — (a) PH_3

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability,



conductivity, or corrosion resistance. That property is specifically associated with PH_3 .

307. Which one of the following reacts with concentrated sulphuric acid

- (a) Au
- (b) Ag
- (c) Pt
- (d) Pb

Answer: B — (b) Ag

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Ag. The choice is consistent with the expected oxidation state and stability of the product.

308. The nitrate of which metal left globule on heating strongly

- (a) $\text{Pb}(\text{NO}_3)_2$
- (b) NaNO_3
- (c) AgNO_3
- (d) $\text{Cu}(\text{NO}_3)_2$

Answer: C — (c) AgNO_3

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives AgNO_3 . The choice is consistent with the expected oxidation state and stability of the product.

309. During extraction of silver, which of the following is formed

- (a) $\text{Na}[\text{Ag}(\text{CN})_2]$
- (b) $\text{Na}_2[\text{Ag}(\text{CN})_2]$
- (c) $\text{Na}_4[\text{Ag}(\text{CN})_2]$
- (d) None of these

Answer: A — (a) $\text{Na}[\text{Ag}(\text{CN})_2]$

Chemical reason: Silver chemistry is governed strongly by low-solubility halides and complex formation with ligands such as NH_3 or $\text{S}_2\text{O}_3^{2-}$. Under the stated conditions this leads to $\text{Na}[\text{Ag}(\text{CN})_2]$.

310. Colourless solutions of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn Blue

- (a) KNO_3
- (b) AgNO_3
- (c) $\text{Zn}(\text{NO}_3)_2$
- (d) ZnSO_4

Answer: B — (b) AgNO_3

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in AgNO_3 has the characteristic oxidation state/ligand field that produces the observed colour.



311. Zinc when reacted with excess of NaOH gives

- (a) Zinc hydroxide
- (b) Zinc oxide
- (c) Di sodium zincate
- (d) Sodium zincate

Answer: D — (d) Sodium zincate

Chemical reason: Excess NaOH dissolves amphoteric zinc hydroxide to form soluble zincate species. In older notation the product is written sodium zincate/disodium zincate depending on formulation.

312. Pair of metals which dissolves in NaOH solution

- (a) Al, Cu
- (b) Zn, Hg
- (c) Zn, Cu
- (d) Zn, Al

Answer: D — (d) Zn, Al

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result Zn, Al is consistent with that behaviour.

313. Lucas reagent is

- (a) Anhydrous ZnCl_2 + conc. HCl
- (b) Hydrated ZnCl_2 + dil. HCl
- (c) Conc. HNO_3 + anhydrous ZnCl_2
- (d) Conc. HNO_3 + anhydrous MgCl_2

Answer: A — (a) Anhydrous ZnCl_2 + conc. HCl

Chemical reason: Lucas reagent is concentrated HCl containing anhydrous ZnCl_2 .

ZnCl_2 acts as a Lewis acid and helps convert alcohols to alkyl chlorides.

314. What is the effect of shaking dil. H_2SO_4 with small quantity of anhydrous CuSO_4

- (a) The white solid dissolves to form a colourless solution
- (b) The white solid dissolves to form a green solution
- (c) The white solid turns blue but does not dissolve
- (d) The white solid dissolves to form a blue solution

Answer: D — (d) The white solid dissolves to form a blue solution

Chemical reason: Anhydrous CuSO_4 is white and becomes blue on hydration because water coordinates to Cu^{2+} , forming hydrated copper(II) sulfate species. This colour change is the basis of its use as a test for water.

315. Which metal is electro-deposited on iron surface to prevent rusting

- (a) Cu
- (b) Zn
- (c) Mg
- (d) Pb

Answer: B — (b) Zn

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Zn matches the relevant metallurgical process or alloy property.



316. To prevent corrosion, iron pipes carrying drinking water are covered with zinc. The process involved is

- (a) Photoelectrolysis
- (b) Electroplating
- (c) Galvanization
- (d) Cathodic protection

Answer: C — (c) Galvanization

Chemical reason: Coating iron with zinc is galvanization. The Zn layer protects both as a barrier and sacrificially if the coating is scratched.

317. From aqueous solution of ZnSO_4 , normal zinc carbonate may be precipitated by

- (a) Boiling with CaCO_3
- (b) Adding Na_2CO_3
- (c) Adding NaHCO_3
- (d) Passing CO_2

Answer: C — (c) Adding NaHCO_3

Chemical reason: The answer follows from solubility and complex-ion equilibria. Under the stated reagent conditions the equilibrium favours Adding NaHCO_3 , whereas the alternatives do not show the same dissolution/precipitation behaviour.

318. Which one of the following dissolve in hot concentrated NaOH solution

- (a) Fe
- (b) Zn
- (c) Cu
- (d) Ag

Answer: B — (b) Zn

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result Zn is consistent with that behaviour.

319. Which of the following metal forms an amphoteric oxide

- (a) Ca
- (b) Fe
- (c) Cu
- (d) Zn

Answer: D — (d) Zn

Chemical reason: Zinc forms amphoteric ZnO , which reacts with both acids and strong bases. That amphoteric behaviour distinguishes it from simple strongly basic metal oxides.

320. Reaction of zinc with cold and very dilute nitric acid yields

- (a) $\text{Zn}(\text{NO}_3)_2 + \text{N}_2\text{O}$
- (b) $\text{Zn}(\text{NO}_3)_2 + \text{NO}$
- (c) $\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3$
- (d) $\text{Zn}(\text{NO}_3)_2 + \text{NO}_2$

Answer: C — (c) $\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3$

Chemical reason: Applying the known reaction of the stated transition-metal compound under





these reagents/conditions gives

$\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3$. The choice is consistent with the expected oxidation state and stability of the product.

